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Project: GNS3 OSPF Network Project

Executive Summary

For this project, I built a network in GNS3 that uses OSPF for routing between multiple routers and VPCS hosts. The network was made from the assigned 10.1.0.0/20 address block. I subnetted the network into smaller point-to-point links for router connections and LAN networks for the VPCS hosts.

The network uses multiple OSPF areas. Area 0 is the backbone area and connects the main routers together. Other areas connect to Area 0 through routers that act as Area Border Routers. This allows the routers to share routing information automatically instead of needing static routes everywhere. The project also includes route summarization and one external connected network that is redistributed into OSPF through R3.

I then tested the network using ping, trace, and OSPF verification commands. The ping and trace results show that hosts on different subnets can reach each other through the routed network. The OSPF neighbor output also shows that routers formed neighbor relationships correctly. The full router configurations and verification outputs are included in the appendix.

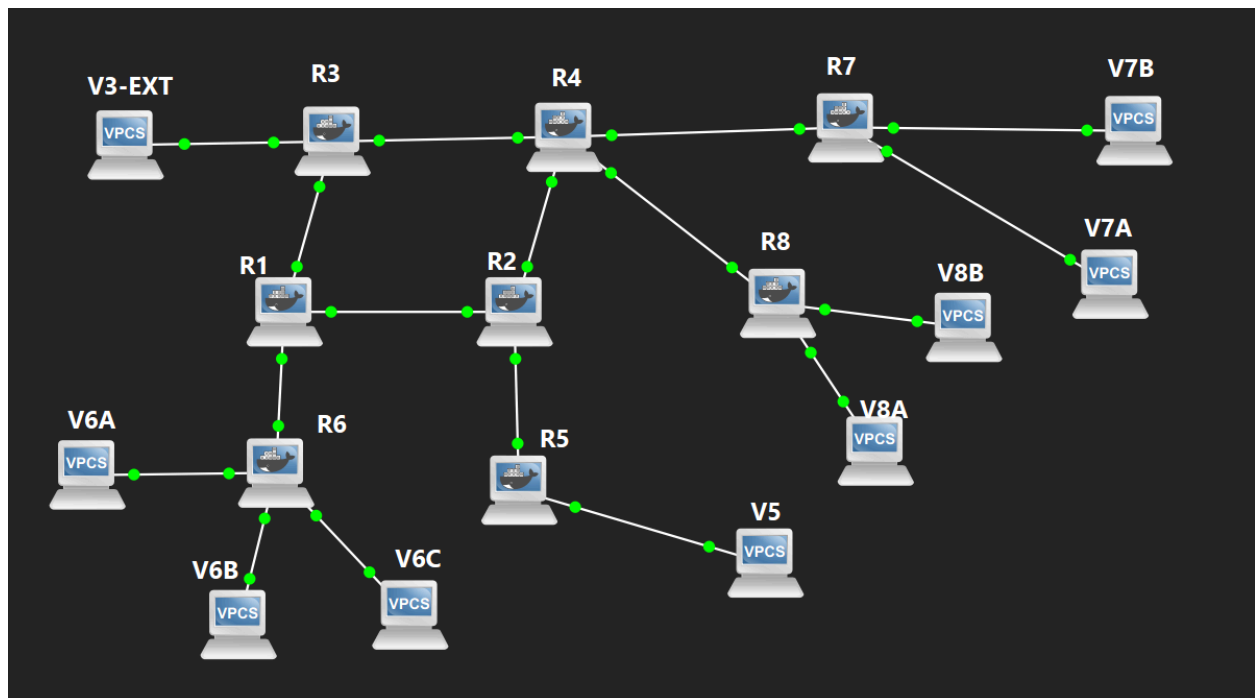
Unexpected Discoveries:

I learned about OSPF in my computer networks class, but it was mostly theoretical knowledge. OSPF is a routing algorithm in the network layer which uses link state advertisements to find shortest paths. But being able to build and use in GNS3 made me better understand how OSPF areas based on my network topology connect to each other which made me have a better grasp at this concept.

Network Topology

The topology was built in GNS3 using 8 routers and multiple VPCS hosts. The routers are connected together using point-to-point links. The VPCS hosts are connected to router LAN interfaces.

The main routers in the backbone are R1, R2, R3, and R4. Other routers connect to the backbone through OSPF areas. R5 is connected through R2, R6 is connected through R1, and R7 and R8 are connected through R4.



OSPF Area Design

OSPF Area	Routers
Area 0	R1, R2, R3, R4
Area 1	R2, R5
Area 2	R4, R7, R8
Area 3	R1, R6

Subnetting Description

The assigned base network was: 10.1.0.0/20

Point-to-Point Router Links

Link	Network	Interface IPs
R1 to R2	10.1.10.0/30	R1 10.1.10.1, R2 10.1.10.2
R1 to R3	10.1.10.4/30	R1 10.1.10.5, R3 10.1.10.6
R2 to R4	10.1.10.8/30	R2 10.1.10.9, R4 10.1.10.10
R3 to R4	10.1.10.12/30	R3 10.1.10.13, R4 10.1.10.14
R2 to R5	10.1.10.16/30	R2 10.1.10.17, R5 10.1.10.18
R1 to R6	10.1.10.20/30	R1 10.1.10.21, R6 10.1.10.22
R4 to R7	10.1.10.24/30	R4 10.1.10.25, R7 10.1.10.26
R4 to R8	10.1.10.28/30	R4 10.1.10.29, R8 10.1.10.30

LAN Networks

Device	Network	Gateway
V5	10.1.0.0/25	10.1.0.1
V6A	10.1.1.0/24	10.1.1.1
V6B	10.1.2.0/24	10.1.2.1
V6C	10.1.4.0/23	10.1.4.1
V7A	10.1.6.0/24	10.1.6.1
V7B	10.1.7.0/24	10.1.7.1
V8A	10.1.8.0/24	10.1.8.1

V8B	10.1.9.0/24	10.1.9.1
V3EXT	10.1.11.0/24	10.1.11.1

Connectivity Testing Example (V6A):

The first thing that came

Hop	Address
1	10.1.1.1
2	10.1.10.21
3	10.1.10.2
4	10.1.10.10
5	10.1.10.26
6	10.1.7.10

OSPF Verification on R1:

From the neighbor test we can see that OSPF is working between these routers.

Neighbor Router ID	Neighbor Address	State
2.2.2.2	10.1.10.2	Full/Backup
3.3.3.3	10.1.10.6	Full/Backup
6.6.6.6	10.1.10.22	Full/Backup

Appendix

Appendix A: Router Configs:

R1

```

conf t
hostname R1
interface eth0
ip address 10.1.10.1/30
no shutdown
exit
interface eth1
ip address 10.1.10.5/30

```

```
no shutdown
exit
interface eth2
ip address 10.1.10.21/30
no shutdown
exit
router ospf
ospf router-id 1.1.1.1
network 10.1.10.0/30 area 0
network 10.1.10.4/30 area 0
network 10.1.10.20/30 area 3
area 3 range 10.1.1.0/24
area 3 range 10.1.2.0/24
area 3 range 10.1.4.0/23
end
write memory
```

R2

```
conf t
hostname R2
interface eth0
ip address 10.1.10.2/30
no shutdown
exit
interface eth1
ip address 10.1.10.9/30
no shutdown
exit
interface eth2
ip address 10.1.10.17/30
no shutdown
exit
router ospf
ospf router-id 2.2.2.2
network 10.1.10.0/30 area 0
network 10.1.10.8/30 area 0
network 10.1.10.16/30 area 1
area 1 range 10.1.0.0/25
end
write memory
```

R3

```
conf t
hostname R3
interface eth0
ip address 10.1.10.6/30
no shutdown
exit
interface eth1
ip address 10.1.10.13/30
no shutdown
exit
interface eth2
ip address 10.1.11.1/24
no shutdown
exit
router ospf
ospf router-id 3.3.3.3
network 10.1.10.4/30 area 0
network 10.1.10.12/30 area 0
redistribute connected
end
write memory
```

R4

```
conf t
hostname R4
interface eth0
ip address 10.1.10.10/30
no shutdown
exit
interface eth1
ip address 10.1.10.14/30
no shutdown
exit
interface eth2
ip address 10.1.10.25/30
no shutdown
exit
interface eth3
ip address 10.1.10.29/30
```

```
no shutdown
exit
router ospf
ospf router-id 4.4.4.4
network 10.1.10.8/30 area 0
network 10.1.10.12/30 area 0
network 10.1.10.24/30 area 2
network 10.1.10.28/30 area 2
area 2 range 10.1.6.0/23
area 2 range 10.1.8.0/23
end
write memory
```

R5

```
conf t
hostname R5
interface eth0
ip address 10.1.10.18/30
no shutdown
exit
interface eth1
ip address 10.1.0.1/25
no shutdown
exit
router ospf
ospf router-id 5.5.5.5
network 10.1.10.16/30 area 1
network 10.1.0.0/25 area 1
end
write memory
```

R6

```
conf t
hostname R6
interface eth0
ip address 10.1.10.22/30
no shutdown
exit
interface eth1
ip address 10.1.1.1/24
no shutdown
exit
```

```
interface eth2
ip address 10.1.2.1/24
no shutdown
exit
interface eth3
ip address 10.1.4.1/23
no shutdown
exit
router ospf
ospf router-id 6.6.6.6
network 10.1.10.20/30 area 3
network 10.1.1.0/24 area 3
network 10.1.2.0/24 area 3
network 10.1.4.0/23 area 3
end
write memory
```

R7

```
conf t
hostname R7
interface eth0
ip address 10.1.10.26/30
no shutdown
exit
interface eth1
ip address 10.1.6.1/24
no shutdown
exit
interface eth2
ip address 10.1.7.1/24
no shutdown
exit
router ospf
ospf router-id 7.7.7.7
network 10.1.10.24/30 area 2
network 10.1.6.0/24 area 2
network 10.1.7.0/24 area 2
end
write memory
```

R8

```
conf t
```



```
hostname R8
interface eth0
ip address 10.1.10.30/30
no shutdown
exit
interface eth1
ip address 10.1.8.1/24
no shutdown
exit
interface eth2
ip address 10.1.9.1/24
no shutdown
exit
router ospf
ospf router-id 8.8.8.8
network 10.1.10.28/30 area 2
network 10.1.8.0/24 area 2
network 10.1.9.0/24 area 2
end
write memory
```

Appendix B: Traceroute Outputs

V5 Traceroute Outputs

V5 to V6A

```
V5> trace 10.1.1.10
trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop
 1 10.1.0.1 0.256 ms 0.354 ms 0.176 ms
 2 10.1.10.17 0.198 ms 0.189 ms 0.185 ms
 3 10.1.10.1 0.334 ms 0.242 ms 0.262 ms
 4 10.1.10.22 0.600 ms 0.747 ms 0.389 ms
 5 *10.1.1.10 0.437 ms (ICMP type:3, code:3, Destination port unreachable)
```

V5 to V6B

```
V5> trace 10.1.2.10
trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop
 1 10.1.0.1 0.217 ms 0.133 ms 0.107 ms
```

2 10.1.10.17 0.257 ms 0.208 ms 0.204 ms
3 10.1.10.1 0.267 ms 0.289 ms 0.208 ms
4 10.1.10.22 0.369 ms 0.286 ms 0.385 ms
5 *10.1.2.10 0.811 ms (ICMP type:3, code:3, Destination port unreachable)

V5 to V6C

V5> trace 10.1.4.10
trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.196 ms 0.182 ms 0.550 ms
2 10.1.10.17 0.239 ms 0.271 ms 0.361 ms
3 10.1.10.1 0.532 ms 0.219 ms 0.317 ms
4 10.1.10.22 0.450 ms 0.270 ms 0.363 ms
5 *10.1.4.10 0.750 ms (ICMP type:3, code:3, Destination port unreachable)

V5 to V7A

V5> trace 10.1.6.10
trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.281 ms 0.127 ms 0.213 ms
2 10.1.10.17 0.322 ms 0.221 ms 0.263 ms
3 10.1.10.10 0.370 ms 0.505 ms 0.294 ms
4 10.1.10.26 0.359 ms 0.301 ms 0.554 ms
5 *10.1.6.10 0.850 ms (ICMP type:3, code:3, Destination port unreachable)

V5 to V7B

V5> trace 10.1.7.10
trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.396 ms 0.128 ms 0.191 ms
2 10.1.10.17 0.246 ms 0.239 ms 0.178 ms
3 10.1.10.10 0.457 ms 0.433 ms 0.413 ms
4 10.1.10.26 0.296 ms 0.361 ms 0.543 ms
5 *10.1.7.10 0.711 ms (ICMP type:3, code:3, Destination port unreachable)

V5 to V8A

V5> trace 10.1.8.10
trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.267 ms 0.175 ms 0.121 ms

```
2 10.1.10.17 0.332 ms 0.173 ms 0.293 ms
3 10.1.10.10 0.251 ms 0.276 ms 0.242 ms
4 10.1.10.30 0.410 ms 0.393 ms 0.375 ms
5 *10.1.8.10 0.801 ms (ICMP type:3, code:3, Destination port unreachable)
```

V5 to V8B

```
V5> trace 10.1.9.10
trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.246 ms 0.260 ms 0.130 ms
2 10.1.10.17 0.374 ms 0.267 ms 0.164 ms
3 10.1.10.10 0.457 ms 0.235 ms 0.300 ms
4 10.1.10.30 0.559 ms 0.290 ms 0.297 ms
5 *10.1.9.10 1.045 ms (ICMP type:3, code:3, Destination port unreachable)
```

V5 to V3EXT

```
V5> trace 10.1.11.10
trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop
1 10.1.0.1 0.350 ms 0.180 ms 0.163 ms
2 10.1.10.17 0.246 ms 0.322 ms 0.157 ms
3 10.1.10.1 0.285 ms 0.289 ms 0.305 ms
4 10.1.10.13 0.549 ms 0.316 ms 0.300 ms
5 *10.1.11.10 1.031 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A Traceroute Outputs

V6A to V5

```
V6A> trace 10.1.0.10
trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop
1 10.1.1.1 0.313 ms 0.185 ms 0.167 ms
2 10.1.10.21 0.257 ms 0.185 ms 0.218 ms
3 10.1.10.2 0.527 ms 0.271 ms 0.373 ms
4 10.1.10.18 0.496 ms 0.311 ms 0.337 ms
5 *10.1.0.10 1.901 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V6B

V6A> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.1.1    0.185 ms 0.188 ms 0.154 ms
2  *10.1.2.10  2.216 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V6C

V6A> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.1.1    0.225 ms 0.114 ms 0.154 ms
2  *10.1.4.10  2.517 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V7A

V6A> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.1.1    0.253 ms 0.161 ms 0.115 ms
2  10.1.10.21  0.224 ms 0.166 ms 0.282 ms
3  10.1.10.2   0.265 ms 0.252 ms 0.249 ms
4  10.1.10.10  0.321 ms 0.323 ms 0.282 ms
5  10.1.10.26  0.340 ms 0.334 ms 0.365 ms
6  *10.1.6.10  3.139 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V7B

V6A> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.1.1    0.235 ms 0.151 ms 0.188 ms
2  10.1.10.21  0.272 ms 0.188 ms 0.293 ms
3  10.1.10.6   0.267 ms 0.230 ms 0.515 ms
4  10.1.10.14  0.391 ms 0.359 ms 0.327 ms
5  10.1.10.26  0.428 ms 0.380 ms 0.317 ms
6  *10.1.7.10  2.812 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V8A

V6A> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.1.1    0.267 ms 0.182 ms 0.169 ms
2  10.1.10.21  0.243 ms 0.203 ms 0.207 ms
```

```
3 10.1.10.2 0.443 ms 0.239 ms 0.281 ms
4 10.1.10.10 0.469 ms 0.469 ms 0.514 ms
5 10.1.10.30 0.377 ms 0.707 ms 0.409 ms
6 *10.1.8.10 2.597 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V8B

V6A> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.1.1 0.270 ms 0.142 ms 0.498 ms
2 10.1.10.21 0.246 ms 0.251 ms 0.296 ms
3 10.1.10.6 0.297 ms 0.373 ms 0.370 ms
4 10.1.10.14 0.270 ms 0.296 ms 0.261 ms
5 10.1.10.30 0.687 ms 0.389 ms 0.410 ms
6 *10.1.9.10 2.216 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6A to V3EXT

V6A> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.1.1 0.185 ms 0.201 ms 0.129 ms
2 10.1.10.21 0.333 ms 0.269 ms 0.197 ms
3 10.1.10.6 0.314 ms 0.280 ms 0.228 ms
4 *10.1.11.10 2.346 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B Traceroute Outputs

V6B to V5

V6B> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1 0.283 ms 0.270 ms 0.217 ms
2 10.1.10.21 0.474 ms 0.225 ms 0.242 ms
3 10.1.10.2 0.339 ms 0.265 ms 0.260 ms
4 10.1.10.18 0.358 ms 0.458 ms 0.288 ms
5 *10.1.0.10 2.203 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V6A

V6B> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1    0.177 ms 0.244 ms 0.124 ms
2 *10.1.1.10  2.057 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V6C

V6B> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1    0.250 ms 0.188 ms 0.189 ms
2 *10.1.4.10  2.077 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V7A

V6B> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1    0.224 ms 0.126 ms 0.217 ms
2 10.1.10.21  0.261 ms 0.201 ms 0.412 ms
3 10.1.10.2   0.334 ms 0.212 ms 0.265 ms
4 10.1.10.10  0.503 ms 0.310 ms 0.307 ms
5 10.1.10.26  0.342 ms 0.430 ms 0.318 ms
6 *10.1.6.10  3.017 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V7B

V6B> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1    0.400 ms 0.195 ms 0.265 ms
2 10.1.10.21  0.228 ms 0.280 ms 0.293 ms
3 10.1.10.2   0.332 ms 0.200 ms 0.212 ms
4 10.1.10.10  0.296 ms 0.398 ms 0.409 ms
5 10.1.10.26  0.381 ms 0.457 ms 0.292 ms
6 *10.1.7.10  2.533 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V8A

V6B> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1    0.288 ms 0.262 ms 0.144 ms
2 10.1.10.21  0.210 ms 0.177 ms 0.269 ms
```

```
3 10.1.10.6 0.341 ms 0.248 ms 0.292 ms
4 10.1.10.14 0.414 ms 0.331 ms 0.435 ms
5 10.1.10.30 0.353 ms 0.327 ms 0.297 ms
6 *10.1.8.10 2.150 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V8B

V6B> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1 0.254 ms 0.166 ms 0.143 ms
2 10.1.10.21 0.210 ms 0.378 ms 0.376 ms
3 10.1.10.6 0.245 ms 0.274 ms 0.259 ms
4 10.1.10.14 0.520 ms 0.452 ms 0.306 ms
5 10.1.10.30 0.474 ms 0.380 ms 0.810 ms
6 *10.1.9.10 5.228 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6B to V3EXT

V6B> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.2.1 0.174 ms 0.134 ms 0.188 ms
2 10.1.10.21 0.191 ms 0.233 ms 0.257 ms
3 10.1.10.6 0.257 ms 0.356 ms 0.244 ms
4 *10.1.11.10 1.499 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C Traceroute Outputs

V6C to V5

V6C> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1 0.271 ms 0.174 ms 0.234 ms
2 10.1.10.21 1.033 ms 0.169 ms 0.226 ms
3 10.1.10.2 0.386 ms 0.283 ms 0.254 ms
4 10.1.10.18 0.308 ms 0.534 ms 0.323 ms
5 *10.1.0.10 0.490 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V6A

V6C> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1    0.197 ms 0.142 ms 0.262 ms
2 *10.1.1.10  0.528 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V6B

V6C> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1    0.340 ms 0.255 ms 0.234 ms
2 *10.1.2.10  0.328 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V7A

V6C> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1    0.391 ms 0.127 ms 0.134 ms
2 10.1.10.21  0.317 ms 0.201 ms 0.244 ms
3 10.1.10.6   0.365 ms 0.266 ms 0.248 ms
4 10.1.10.14  0.286 ms 0.356 ms 0.296 ms
5 10.1.10.26  0.495 ms 0.400 ms 0.381 ms
6 *10.1.6.10  0.697 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V7B

V6C> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1    0.194 ms 0.113 ms 0.140 ms
2 10.1.10.21  0.262 ms 0.215 ms 0.266 ms
3 10.1.10.2   0.385 ms 0.284 ms 0.352 ms
4 10.1.10.10  0.475 ms 0.438 ms 0.316 ms
5 10.1.10.26  0.426 ms 0.370 ms 0.378 ms
6 *10.1.7.10  0.969 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V8A

V6C> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1    0.207 ms 0.109 ms 0.174 ms
2 10.1.10.21  0.226 ms 0.182 ms 0.229 ms
3 10.1.10.2   0.363 ms 0.237 ms 0.270 ms
```



```
4 10.1.10.10 1.056 ms 0.328 ms 0.295 ms
5 10.1.10.30 0.416 ms 0.325 ms 0.429 ms
6 *10.1.8.10 0.702 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V8B

V6C> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1 0.176 ms 0.137 ms 0.175 ms
2 10.1.10.21 0.416 ms 0.268 ms 0.217 ms
3 10.1.10.2 0.258 ms 0.290 ms 0.240 ms
4 10.1.10.10 0.276 ms 0.296 ms 0.316 ms
5 10.1.10.30 0.432 ms 0.710 ms 0.357 ms
6 *10.1.9.10 0.635 ms (ICMP type:3, code:3, Destination port unreachable)
```

V6C to V3EXT

V6C> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.4.1 0.352 ms 0.166 ms 0.131 ms
2 10.1.10.21 0.279 ms 0.196 ms 0.178 ms
3 10.1.10.6 0.305 ms 0.422 ms 0.393 ms
4 *10.1.11.10 0.420 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A Traceroute Outputs

V7A to V5

V7A> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1 0.412 ms 0.124 ms 0.179 ms
2 10.1.10.25 0.330 ms 0.186 ms 0.297 ms
3 10.1.10.9 0.364 ms 0.221 ms 0.251 ms
4 10.1.10.18 0.355 ms 0.257 ms 0.427 ms
5 *10.1.0.10 0.388 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V6A

V7A> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.182 ms 0.240 ms 0.169 ms
2 10.1.10.25  0.284 ms 0.206 ms 0.262 ms
3 10.1.10.9   0.356 ms 0.248 ms 0.230 ms
4 10.1.10.1   0.466 ms 0.294 ms 0.354 ms
5 10.1.10.22  0.473 ms 0.300 ms 0.308 ms
6 *10.1.1.10  2.371 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V6B

V7A> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.291 ms 0.161 ms 0.146 ms
2 10.1.10.25  0.270 ms 0.167 ms 0.219 ms
3 10.1.10.9   0.278 ms 0.316 ms 0.259 ms
4 10.1.10.1   0.357 ms 0.480 ms 0.348 ms
5 10.1.10.22  0.409 ms 0.463 ms 0.416 ms
6 *10.1.2.10  1.949 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V6C

V7A> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.276 ms 0.117 ms 0.134 ms
2 10.1.10.25  0.229 ms 0.185 ms 0.543 ms
3 10.1.10.13  0.280 ms 0.401 ms 0.303 ms
4 10.1.10.5   0.382 ms 0.315 ms 0.313 ms
5 10.1.10.22  0.553 ms 0.373 ms 0.640 ms
6 *10.1.4.10  0.549 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V7B

V7A> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.332 ms 0.131 ms 0.137 ms
2 *10.1.7.10  2.105 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V8A

V7A> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.280 ms 0.188 ms 0.147 ms
2 10.1.10.25  0.463 ms 0.344 ms 0.211 ms
3 10.1.10.30  0.297 ms 0.269 ms 0.325 ms
4 *10.1.8.10  1.820 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V8B

V7A> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.236 ms 0.110 ms 0.165 ms
2 10.1.10.25  0.201 ms 0.180 ms 0.371 ms
3 10.1.10.30  0.272 ms 0.235 ms 0.360 ms
4 *10.1.9.10  1.699 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7A to V3EXT

V7A> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.6.1    0.278 ms 0.153 ms 0.185 ms
2 10.1.10.25  0.289 ms 0.745 ms 0.240 ms
3 10.1.10.13  0.382 ms 0.236 ms 0.520 ms
4 *10.1.11.10 2.163 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B Traceroute Outputs

V7B to V5

V7B> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1    0.231 ms 0.123 ms 0.120 ms
2 10.1.10.25  0.255 ms 0.182 ms 0.145 ms
3 10.1.10.9   0.166 ms 0.223 ms 0.234 ms
4 10.1.10.18  0.363 ms 0.370 ms 0.213 ms
5 *10.1.0.10  1.792 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V6A

V7B> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1    0.168 ms 0.159 ms 0.100 ms
2 10.1.10.25  0.181 ms 0.135 ms 0.142 ms
3 10.1.10.13  0.240 ms 0.211 ms 0.143 ms
```

```
4 10.1.10.5 0.243 ms 0.228 ms 0.178 ms
5 10.1.10.22 0.338 ms 0.232 ms 0.291 ms
6 *10.1.1.10 1.786 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V6B

V7B> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1 0.257 ms 0.075 ms 0.155 ms
2 10.1.10.25 0.208 ms 0.160 ms 0.182 ms
3 10.1.10.9 0.148 ms 0.193 ms 0.184 ms
4 10.1.10.1 0.256 ms 0.246 ms 0.262 ms
5 10.1.10.22 0.243 ms 0.205 ms 0.261 ms
6 *10.1.2.10 1.896 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V6C

V7B> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1 0.236 ms 0.118 ms 0.157 ms
2 10.1.10.25 0.137 ms 0.124 ms 0.193 ms
3 10.1.10.9 0.219 ms 0.188 ms 0.193 ms
4 10.1.10.1 0.222 ms 0.190 ms 0.240 ms
5 10.1.10.22 0.293 ms 0.218 ms 0.361 ms
6 *10.1.4.10 1.730 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V7A

V7B> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1 0.146 ms 0.141 ms 0.142 ms
2 *10.1.6.10 1.830 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V8A

V7B> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1 10.1.7.1 0.173 ms 0.094 ms 0.101 ms
2 10.1.10.25 0.206 ms 0.144 ms 0.185 ms
3 10.1.10.30 0.230 ms 0.185 ms 0.184 ms
4 *10.1.8.10 1.943 ms (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V8B

V7B> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.7.1    0.198 ms  0.146 ms  0.099 ms
2  10.1.10.25  0.133 ms  0.130 ms  0.207 ms
3  10.1.10.30  0.197 ms  0.170 ms  0.159 ms
4  *10.1.9.10  1.659 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V7B to V3EXT

V7B> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.7.1    0.131 ms  0.146 ms  0.158 ms
2  10.1.10.25  0.176 ms  0.105 ms  0.184 ms
3  10.1.10.13  0.231 ms  0.151 ms  0.202 ms
4  *10.1.11.10 1.787 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A Traceroute Outputs

V8A to V6A

V8A> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.213 ms  0.133 ms  0.137 ms
2  10.1.10.29  0.252 ms  0.165 ms  0.153 ms
3  10.1.10.9   0.196 ms  0.181 ms  0.178 ms
4  10.1.10.1   0.242 ms  0.221 ms  0.268 ms
5  10.1.10.22  0.496 ms  0.237 ms  0.264 ms
6  *10.1.1.10  1.981 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V6B

V8A> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.162 ms  0.119 ms  0.115 ms
2  10.1.10.29  0.183 ms  0.183 ms  0.157 ms
3  10.1.10.13  0.167 ms  0.215 ms  0.244 ms
4  10.1.10.5   0.333 ms  0.195 ms  0.266 ms
5  10.1.10.22  0.328 ms  0.292 ms  0.284 ms
6  *10.1.2.10  1.957 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V6C

V8A> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.110 ms  0.092 ms  0.090 ms
2  10.1.10.29  0.163 ms  0.197 ms  0.223 ms
3  10.1.10.9   0.169 ms  0.206 ms  0.209 ms
4  10.1.10.1   0.291 ms  0.212 ms  0.243 ms
5  10.1.10.22  0.245 ms  0.234 ms  0.313 ms
6  *10.1.4.10  1.942 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V7A

V8A> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.169 ms  0.138 ms  0.118 ms
2  10.1.10.29  0.158 ms  0.116 ms  0.192 ms
3  10.1.10.26  0.243 ms  0.154 ms  0.182 ms
4  *10.1.6.10  1.610 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V7B

V8A> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.158 ms  0.122 ms  0.180 ms
2  10.1.10.29  0.187 ms  0.116 ms  0.194 ms
3  10.1.10.26  0.233 ms  0.286 ms  0.197 ms
4  *10.1.7.10  2.068 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V8B

V8A> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.210 ms  0.090 ms  0.117 ms
2  *10.1.9.10  1.720 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8A to V3EXT

V8A> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.8.1    0.167 ms  0.096 ms  0.255 ms
2  10.1.10.29  0.137 ms  0.141 ms  0.190 ms
3  10.1.10.13  0.205 ms  0.194 ms  0.191 ms
4  *10.1.11.10 1.453 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8B Traceroute Outputs

V8B to V5

V8B> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

1	10.1.9.1	0.247 ms	0.148 ms	0.108 ms
2	10.1.10.29	0.163 ms	0.190 ms	0.132 ms
3	10.1.10.9	0.275 ms	0.220 ms	0.215 ms
4	10.1.10.18	0.229 ms	0.262 ms	0.263 ms
5	*10.1.0.10	0.446 ms	(ICMP type:3, code:3, Destination port unreachable)	

V8B to V6A

V8B> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

1	10.1.9.1	0.246 ms	0.182 ms	0.099 ms
2	10.1.10.29	0.192 ms	0.145 ms	0.110 ms
3	10.1.10.13	0.202 ms	0.164 ms	0.234 ms
4	10.1.10.5	0.282 ms	0.265 ms	0.200 ms
5	10.1.10.22	0.340 ms	0.183 ms	0.219 ms
6	*10.1.1.10	0.251 ms	(ICMP type:3, code:3, Destination port unreachable)	

V8B to V6B

V8B> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

1	10.1.9.1	0.243 ms	0.121 ms	0.174 ms
2	10.1.10.29	0.171 ms	0.126 ms	0.197 ms
3	10.1.10.13	0.211 ms	0.184 ms	0.174 ms
4	10.1.10.5	0.248 ms	0.166 ms	0.158 ms
5	10.1.10.22	0.246 ms	0.242 ms	0.214 ms
6	*10.1.2.10	0.326 ms	(ICMP type:3, code:3, Destination port unreachable)	

V8B to V6C

V8B> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

1	10.1.9.1	0.136 ms	0.117 ms	0.138 ms
2	10.1.10.29	0.119 ms	0.191 ms	0.120 ms
3	10.1.10.9	0.155 ms	0.179 ms	0.199 ms
4	10.1.10.1	0.205 ms	0.191 ms	0.187 ms
5	10.1.10.22	0.326 ms	0.208 ms	0.251 ms
6	*10.1.4.10	0.468 ms	(ICMP type:3, code:3, Destination port unreachable)	

V8B to V7A

V8B> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.9.1    0.148 ms  0.092 ms  0.137 ms
2  10.1.10.29  0.159 ms  0.156 ms  0.188 ms
3  10.1.10.26  0.201 ms  0.190 ms  0.236 ms
4  *10.1.6.10  0.345 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8B to V7B

V8B> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.9.1    0.161 ms  0.112 ms  0.091 ms
2  10.1.10.29  0.210 ms  0.142 ms  0.205 ms
3  10.1.10.26  0.197 ms  0.194 ms  0.191 ms
4  *10.1.7.10  0.285 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8B to V8A

V8B> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.9.1    0.221 ms  0.100 ms  0.099 ms
2  *10.1.8.10  0.167 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V8B to V3EXT

V8B> trace 10.1.11.10

trace to 10.1.11.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.9.1    0.156 ms  0.124 ms  0.196 ms
2  10.1.10.29  0.167 ms  0.117 ms  0.166 ms
3  10.1.10.13  0.197 ms  0.182 ms  0.169 ms
4  *10.1.11.10 0.320 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V3-EXT Traceroute Outputs

V3-EXT to V5

V3-EXT> trace 10.1.0.10

trace to 10.1.0.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.11.1    0.226 ms  0.096 ms  0.103 ms
2  10.1.10.14   0.146 ms  0.108 ms  0.114 ms
3  10.1.10.2    0.237 ms  0.152 ms  0.166 ms
4  10.1.10.18   0.315 ms  0.183 ms  0.149 ms
5  *10.1.0.10  1.752 ms  (ICMP type:3, code:3, Destination port unreachable)
```


V3-EXT to V6A

V3-EXT> trace 10.1.1.10

trace to 10.1.1.10, 8 hops max, press Ctrl+C to stop

1	10.1.11.1	0.125 ms	0.153 ms	0.130 ms
2	10.1.10.5	0.183 ms	0.183 ms	0.129 ms
3	10.1.10.22	0.252 ms	0.237 ms	0.180 ms
4	*10.1.1.10	1.440 ms	(ICMP type:3, code:3, Destination port unreachable)	

V3-EXT to V6B

V3-EXT> trace 10.1.2.10

trace to 10.1.2.10, 8 hops max, press Ctrl+C to stop

1	10.1.11.1	0.175 ms	0.130 ms	0.091 ms
2	10.1.10.5	0.206 ms	0.125 ms	0.239 ms
3	10.1.10.22	0.227 ms	0.153 ms	0.197 ms
4	*10.1.2.10	1.982 ms	(ICMP type:3, code:3, Destination port unreachable)	

V3-EXT to V6C

V3-EXT> trace 10.1.4.10

trace to 10.1.4.10, 8 hops max, press Ctrl+C to stop

1	10.1.11.1	0.236 ms	0.133 ms	0.096 ms
2	10.1.10.5	0.177 ms	0.168 ms	0.136 ms
3	10.1.10.22	0.162 ms	0.205 ms	0.180 ms
4	*10.1.4.10	1.749 ms	(ICMP type:3, code:3, Destination port unreachable)	

V3-EXT to V7A

V3-EXT> trace 10.1.6.10

trace to 10.1.6.10, 8 hops max, press Ctrl+C to stop

1	10.1.11.1	0.136 ms	0.098 ms	0.092 ms
2	10.1.10.14	0.155 ms	0.128 ms	0.129 ms
3	10.1.10.26	0.200 ms	0.181 ms	0.243 ms
4	*10.1.6.10	1.731 ms	(ICMP type:3, code:3, Destination port unreachable)	

V3-EXT to V7B

V3-EXT> trace 10.1.7.10

trace to 10.1.7.10, 8 hops max, press Ctrl+C to stop

1	10.1.11.1	0.224 ms	0.089 ms	0.217 ms
2	10.1.10.14	0.190 ms	0.139 ms	0.222 ms
3	10.1.10.26	0.170 ms	0.216 ms	0.168 ms
4	*10.1.7.10	1.782 ms	(ICMP type:3, code:3, Destination port unreachable)	

V3-EXT to V8A

V3-EXT> trace 10.1.8.10

trace to 10.1.8.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.11.1    0.151 ms  0.118 ms  0.093 ms
2  10.1.10.14   0.201 ms  0.179 ms  0.166 ms
3  10.1.10.30   0.285 ms  0.217 ms  0.269 ms
4  *10.1.8.10  1.343 ms  (ICMP type:3, code:3, Destination port unreachable)
```

V3-EXT to V8B

V3-EXT> trace 10.1.9.10

trace to 10.1.9.10, 8 hops max, press Ctrl+C to stop

```
1  10.1.11.1    0.191 ms  0.159 ms  0.094 ms
2  10.1.10.14   0.162 ms  0.164 ms  0.159 ms
3  10.1.10.30   0.205 ms  0.189 ms  0.201 ms
4  *10.1.9.10  1.836 ms  (ICMP type:3, code:3, Destination port unreachable)
```

Appendix C: OSPF Database Outputs

R1 OSPF Database Output

R1# show ip ospf database

Router Link States Area 0.0.0.3

Link ID	ADV Router	Age	Seq#	CkSum	Link count
1.1.1.1	1.1.1.1	1305	0x80000006	0xccec	1
6.6.6.6	6.6.6.6	1332	0x8000000a	0xa5ea	4

Net Link States Area 0.0.0.3

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.22	6.6.6.6	1312	0x80000002	0x2ece

Summary Link States Area 0.0.0.3

Link ID	ADV Router	Age	Seq#	CkSum	Route
---------	------------	-----	------	-------	-------

10.1.0.0	1.1.1.1	1275 0x80000003 0xc065	10.1.0.0/25
10.1.6.0	1.1.1.1	1305 0x80000002 0x79f1	10.1.6.0/23
10.1.8.0	1.1.1.1	1285 0x80000002 0x6306	10.1.8.0/23
10.1.10.0	1.1.1.1	1345 0x80000002 0x3dd1	10.1.10.0/30
10.1.10.4	1.1.1.1	1315 0x80000003 0x13f6	10.1.10.4/30
10.1.10.8	1.1.1.1	1315 0x80000003 0xebe2	10.1.10.8/30
10.1.10.12	1.1.1.1	1335 0x80000002 0xc506	10.1.10.12/30
10.1.10.16	1.1.1.1	1325 0x80000003 0x9b2b	10.1.10.16/30
10.1.10.24	1.1.1.1	1305 0x80000002 0x4e3a	10.1.10.24/30
10.1.10.28	1.1.1.1	1285 0x80000002 0x265e	10.1.10.28/30

ASBR Summary Link States Area 0.0.0.3

Link ID	ADV Router	Age Seq#	CkSum
3.3.3.3	1.1.1.1	1305 0x80000002	0xb360

AS External Link States

Link ID	ADV Router	Age Seq#	CkSum	Route
10.1.11.0	3.3.3.3	1366 0x80000002	0x4754	E2 10.1.11.0/24 [0x0]

R2 OSPF Database Output

R2# show ip ospf database

OSPF Router with ID (2.2.2.2)

Router Link States Area 0.0.0.0

Link ID	ADV Router	Age Seq#	CkSum	Link count
1.1.1.1	1.1.1.1	1511 0x80000008	0x3331	2
2.2.2.2	2.2.2.2	1490 0x80000009	0xb39e	2
3.3.3.3	3.3.3.3	1512 0x80000008	0x8dac	2
4.4.4.4	4.4.4.4	1530 0x80000008	0xb871	2

Net Link States Area 0.0.0.0

Link ID	ADV Router	Age Seq#	CkSum
10.1.10.2	2.2.2.2	1510 0x80000002	0xa68a
10.1.10.6	3.3.3.3	1502 0x80000002	0x9292

10.1.10.10	4.4.4.4	1540 0x80000002 0xa074
10.1.10.14	4.4.4.4	1510 0x80000002 0x9a72

Summary Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.0.0	2.2.2.2	1500	0x80000002	0xa3b6	10.1.0.0/25
10.1.1.0	1.1.1.1	1501	0x80000002	0xb32a	10.1.1.0/24
10.1.2.0	1.1.1.1	1531	0x80000002	0xa834	10.1.2.0/24
10.1.4.0	1.1.1.1	1531	0x80000002	0x8d4e	10.1.4.0/23
10.1.6.0	4.4.4.4	1510	0x80000002	0x1db0	10.1.6.0/23
10.1.8.0	4.4.4.4	1522	0x80000002	0x07c4	10.1.8.0/23
10.1.10.16	2.2.2.2	1510	0x80000003	0x7c7d	10.1.10.16/30
10.1.10.20	1.1.1.1	1531	0x80000003	0x7287	10.1.10.20/30
10.1.10.24	4.4.4.4	1500	0x80000003	0xeff9	10.1.10.24/30
10.1.10.28	4.4.4.4	1540	0x80000003	0xc71e	10.1.10.28/30

Router Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum	Link count
2.2.2.2	2.2.2.2	1530	0x80000006	0x2099	1
5.5.5.5	5.5.5.5	1489	0x80000006	0xe6e7	2

Net Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.18	5.5.5.5	1519	0x80000002	0x64a0

Summary Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.1.0	2.2.2.2	1510	0x80000003	0x940d	10.1.1.0/24
10.1.2.0	2.2.2.2	1510	0x80000003	0x8917	10.1.2.0/24
10.1.4.0	2.2.2.2	1520	0x80000003	0x6e31	10.1.4.0/23
10.1.6.0	2.2.2.2	1520	0x80000002	0x5a44	10.1.6.0/23
10.1.8.0	2.2.2.2	1520	0x80000002	0x4458	10.1.8.0/23
10.1.10.0	2.2.2.2	1540	0x80000002	0x1feb	10.1.10.0/30
10.1.10.4	2.2.2.2	1510	0x80000003	0xf5d8	10.1.10.4/30
10.1.10.8	2.2.2.2	1540	0x80000002	0xce34	10.1.10.8/30
10.1.10.12	2.2.2.2	1500	0x80000002	0xa720	10.1.10.12/30

10.1.10.20	2.2.2.2	1470	0x80000003	0x5569	10.1.10.20/30
10.1.10.24	2.2.2.2	1500	0x80000002	0x2f8c	10.1.10.24/30
10.1.10.28	2.2.2.2	1510	0x80000002	0x07b0	10.1.10.28/30

ASBR Summary Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum
3.3.3.3	2.2.2.2	1530	0x80000002	0x9642

AS External Link States

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.11.0	3.3.3.3	1562	0x80000002	0x4754	E2 10.1.11.0/24 [0x0]

R3 OSPF Database Output

R3# show ip ospf database

OSPF Router with ID (3.3.3.3)

Router Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum	Link count
1.1.1.1	1.1.1.1	1567	0x80000008	0x3331	2
2.2.2.2	2.2.2.2	1548	0x80000009	0xb39e	2
3.3.3.3	3.3.3.3	1566	0x80000008	0x8dac	2
4.4.4.4	4.4.4.4	1586	0x80000008	0xb871	2

Net Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.2	2.2.2.2	1568	0x80000002	0xa68a
10.1.10.6	3.3.3.3	1556	0x80000002	0x9292
10.1.10.10	4.4.4.4	1596	0x80000002	0xa074
10.1.10.14	4.4.4.4	1566	0x80000002	0x9a72

Summary Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum	Route
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10.1.0.0	2.2.2.2	1558 0x80000002 0xa3b6	10.1.0.0/25
10.1.1.0	1.1.1.1	1557 0x80000002 0xb32a	10.1.1.0/24
10.1.2.0	1.1.1.1	1587 0x80000002 0xa834	10.1.2.0/24
10.1.4.0	1.1.1.1	1587 0x80000002 0x8d4e	10.1.4.0/23
10.1.6.0	4.4.4.4	1566 0x80000002 0x1db0	10.1.6.0/23
10.1.8.0	4.4.4.4	1576 0x80000002 0x07c4	10.1.8.0/23
10.1.10.16	2.2.2.2	1568 0x80000003 0x7c7d	10.1.10.16/30
10.1.10.20	1.1.1.1	1587 0x80000003 0x7287	10.1.10.20/30
10.1.10.24	4.4.4.4	1556 0x80000003 0xe7f9	10.1.10.24/30
10.1.10.28	4.4.4.4	1596 0x80000003 0xc71e	10.1.10.28/30

AS External Link States

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.11.0	3.3.3.3	1616	0x80000002	0x4754	E2 10.1.11.0/24 [0x0]

R4 OSPF Database Output

R4# show ip ospf database

OSPF Router with ID (4.4.4.4)

Router Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum	Link count
1.1.1.1	1.1.1.1	1721	0x80000008	0x3331	2
2.2.2.2	2.2.2.2	1700	0x80000009	0xb39e	2
3.3.3.3	30	0x80000009	0x8bad	2	
4.4.4.4	38	0x80000009	0xb672	2	

Net Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.2	2.2.2.2	1720	0x80000002	0xa68a
10.1.10.6	3.3.3.3	20	0x80000003	0x9093
10.1.10.10	4.4.4.4	18	0x80000003	0x9e75
10.1.10.14	4.4.4.4	18	0x80000003	0x9873

Summary Link States Area 0.0.0.0

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.0.0	2.2.2.2	1710	0x80000002	0xa3b6	10.1.0.0/25
10.1.1.0	1.1.1.1	1711	0x80000002	0xb32a	10.1.1.0/24
10.1.2.0	1.1.1.1	51	0x80000003	0xa635	10.1.2.0/24
10.1.4.0	1.1.1.1	41	0x80000003	0x8b4f	10.1.4.0/23
10.1.6.0	4.4.4.4	1718	0x80000002	0x1db0	10.1.6.0/23
10.1.8.0	4.4.4.4	8	0x80000003	0x05c5	10.1.8.0/23
10.1.10.16	2.2.2.2	30	0x80000004	0x7a7e	10.1.10.16/30
10.1.10.20	1.1.1.1	41	0x80000004	0x7088	10.1.10.20/30
10.1.10.24	4.4.4.4	1708	0x80000003	0xe9ff	10.1.10.24/30
10.1.10.28	4.4.4.4	28	0x80000004	0xc51f	10.1.10.28/30

Router Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum	Link count
4.4.4.4	4.4.4.4	8	0x8000000b	0x0fd9	2
7.7.7.7	7.7.7.7	1726	0x80000008	0x3796	3
8.8.8.8	8.8.8.8	15	0x80000009	0xb503	3

Net Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.26	7.7.7.7	1696	0x80000002	0x8064
10.1.10.30	8.8.8.8	5	0x80000003	0x6a6d

Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.0.0	4.4.4.4	8	0x80000003	0x66b3	10.1.0.0/25
10.1.1.0	4.4.4.4	8	0x80000003	0x5909	10.1.1.0/24
10.1.2.0	4.4.4.4	1718	0x80000002	0x5012	10.1.2.0/24
10.1.4.0	4.4.4.4	28	0x80000003	0x332d	10.1.4.0/23
10.1.10.0	4.4.4.4	18	0x80000003	0xe1e8	10.1.10.0/30
10.1.10.4	4.4.4.4	8	0x80000003	0xb90d	10.1.10.4/30
10.1.10.8	4.4.4.4	78	0x80000003	0x9069	10.1.10.8/30
10.1.10.12	4.4.4.4	18	0x80000003	0x688d	10.1.10.12/30
10.1.10.16	4.4.4.4	1698	0x80000002	0x4378	10.1.10.16/30
10.1.10.20	4.4.4.4	1698	0x80000002	0x1c64	10.1.10.20/30

ASBR Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum
3.3.3.3	4.4.4.4	28	0x80000003	0x57af

AS External Link States

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.11.0	3.3.3.3	50	0x80000003	0x4555	E2 10.1.11.0/24 [0x0]

R5 OSPF Database Output

R5# show ip ospf database

OSPF Router with ID (5.5.5.5)

Router Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum	Link count
2.2.2.2	2.2.2.2	735	0x80000008	0x1c9b	1
5.5.5.5	5.5.5.5	662	0x80000008	0xe2e9	2

Net Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.18	5.5.5.5	672	0x80000004	0x60a2

Summary Link States Area 0.0.0.1

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.1.0	2.2.2.2	675	0x80000005	0x900f	10.1.1.0/24
10.1.2.0	2.2.2.2	715	0x80000005	0x8519	10.1.2.0/24
10.1.4.0	2.2.2.2	665	0x80000005	0x6a33	10.1.4.0/23
10.1.6.0	2.2.2.2	705	0x80000004	0x5646	10.1.6.0/23
10.1.8.0	2.2.2.2	695	0x80000004	0x405a	10.1.8.0/23
10.1.10.0	2.2.2.2	665	0x80000004	0x1bed	10.1.10.0/30
10.1.10.4	2.2.2.2	705	0x80000005	0xf1da	10.1.10.4/30
10.1.10.8	2.2.2.2	675	0x80000004	0xca36	10.1.10.8/30
10.1.10.12	2.2.2.2	665	0x80000004	0xa322	10.1.10.12/30
10.1.10.20	2.2.2.2	655	0x80000005	0x516b	10.1.10.20/30

10.1.10.24	2.2.2.2	685 0x80000004 0x2b8e	10.1.10.24/30
10.1.10.28	2.2.2.2	645 0x80000004 0x03b2	10.1.10.28/30

ASBR Summary Link States Area 0.0.0.1

Link ID	ADV Router	Age Seq#	CkSum
3.3.3.3	2.2.2.2	695 0x80000004	0x9244

AS External Link States

Link ID	ADV Router	Age Seq#	CkSum	Route
10.1.11.0	3.3.3.3	747 0x80000004	0x4356	E2 10.1.11.0/24 [0x0]

R6 OSPF Database Output

R6# show ip ospf database

OSPF Router with ID (6.6.6.6)

Router Link States Area 0.0.0.3

Link ID	ADV Router	Age Seq#	CkSum	Link count
1.1.1.1	1.1.1.1	755 0x80000008	0xc8ee	1
6.6.6.6	6.6.6.6	751 0x8000000c	0xa1ec	4

Net Link States Area 0.0.0.3

Link ID	ADV Router	Age Seq#	CkSum
10.1.10.22	6.6.6.6	711 0x80000004	0x2ad0

Summary Link States Area 0.0.0.3

Link ID	ADV Router	Age Seq#	CkSum	Route
10.1.0.0	1.1.1.1	685 0x80000005	0xbc67	10.1.0.0/25
10.1.6.0	1.1.1.1	755 0x80000004	0x75f3	10.1.6.0/23
10.1.8.0	1.1.1.1	665 0x80000004	0x5f08	10.1.8.0/23
10.1.10.0	1.1.1.1	745 0x80000004	0x39d3	10.1.10.0/30
10.1.10.4	1.1.1.1	685 0x80000005	0x0ff8	10.1.10.4/30
10.1.10.8	1.1.1.1	715 0x80000005	0xe7e4	10.1.10.8/30
10.1.10.12	1.1.1.1	785 0x80000004	0xc108	10.1.10.12/30

10.1.10.16	1.1.1.1	705	0x80000005	0x972d	10.1.10.16/30
10.1.10.24	1.1.1.1	735	0x80000004	0x4a3c	10.1.10.24/30
10.1.10.28	1.1.1.1	705	0x80000004	0x2260	10.1.10.28/30

ASBR Summary Link States Area 0.0.0.3

Link ID	ADV Router	Age	Seq#	CkSum
3.3.3.3	1.1.1.1	695	0x80000004	0xaf62

AS External Link States

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.11.0	3.3.3.3	796	0x80000004	0x4356	E2 10.1.11.0/24 [0x0]

R7 OSPF Database Output

R7# show ip ospf database

OSPF Router with ID (7.7.7.7)

Router Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum	Link count
4.4.4.4	4.4.4.4	809	0x8000000c	0x0dda	2
7.7.7.7	7.7.7.7	755	0x8000000a	0x3398	3
8.8.8.8	8.8.8.8	786	0x8000000a	0xb304	3

Net Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum
10.1.10.26	7.7.7.7	735	0x80000004	0x7c66
10.1.10.30	8.8.8.8	766	0x80000004	0x686e

Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.0.0	4.4.4.4	769	0x80000004	0x64b4	10.1.0.0/25
10.1.1.0	4.4.4.4	809	0x80000004	0x570a	10.1.1.0/24
10.1.2.0	4.4.4.4	789	0x80000004	0x4c14	10.1.2.0/24

10.1.4.0	4.4.4.4	839 0x80000004 0x312e	10.1.4.0/23
10.1.10.0	4.4.4.4	799 0x80000004 0xdfe9	10.1.10.0/30
10.1.10.4	4.4.4.4	789 0x80000004 0xb70e	10.1.10.4/30
10.1.10.8	4.4.4.4	879 0x80000004 0x8e6a	10.1.10.8/30
10.1.10.12	4.4.4.4	779 0x80000004 0x668e	10.1.10.12/30
10.1.10.16	4.4.4.4	779 0x80000004 0x3f7a	10.1.10.16/30
10.1.10.20	4.4.4.4	759 0x80000004 0x1866	10.1.10.20/30

ASBR Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age Seq#	CkSum
3.3.3.3	4.4.4.4	829 0x80000004	0x55b0

AS External Link States

Link ID	ADV Router	Age Seq#	CkSum	Route
10.1.11.0	3.3.3.3	852 0x80000004	0x4356	E2 10.1.11.0/24 [0x0]

R8 OSPF Database Output

R8# show ip ospf database

OSPF Router with ID (8.8.8.8)

Router Link States Area 0.0.0.2

Link ID	ADV Router	Age Seq#	CkSum	Link count
4.4.4.4	4.4.4.4	849 0x8000000c	0x0dda	2
7.7.7.7	7.7.7.7	797 0x8000000a	0x3398	3
8.8.8.8	8.8.8.8	824 0x8000000a	0xb304	3

Net Link States Area 0.0.0.2

Link ID	ADV Router	Age Seq#	CkSum
10.1.10.26	7.7.7.7	777 0x80000004	0x7c66
10.1.10.30	8.8.8.8	804 0x80000004	0x686e

Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.0.0	4.4.4.4	809	0x80000004	0x64b4	10.1.0.0/25
10.1.1.0	4.4.4.4	849	0x80000004	0x570a	10.1.1.0/24
10.1.2.0	4.4.4.4	829	0x80000004	0x4c14	10.1.2.0/24
10.1.4.0	4.4.4.4	879	0x80000004	0x312e	10.1.4.0/23
10.1.10.0	4.4.4.4	839	0x80000004	0xdfe9	10.1.10.0/30
10.1.10.4	4.4.4.4	829	0x80000004	0xb70e	10.1.10.4/30
10.1.10.8	4.4.4.4	919	0x80000004	0x8e6a	10.1.10.8/30
10.1.10.12	4.4.4.4	819	0x80000004	0x668e	10.1.10.12/30
10.1.10.16	4.4.4.4	819	0x80000004	0x3f7a	10.1.10.16/30
10.1.10.20	4.4.4.4	799	0x80000004	0x1866	10.1.10.20/30

ASBR Summary Link States Area 0.0.0.2

Link ID	ADV Router	Age	Seq#	CkSum
3.3.3.3	4.4.4.4	869	0x80000004	0x55b0

AS External Link States

Link ID	ADV Router	Age	Seq#	CkSum	Route
10.1.11.0	3.3.3.3	892	0x80000004	0x4356	E2 10.1.11.0/24 [0x0]